

Juan David Muñoz-Bolaños

R&D Optical Engineer | Adaptive Optics & Machine Vision

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SUMMARY

R&D Engineer with a dual background in **Physics Engineering** and **Advanced Microscopy**. Expert in bridging the gap between optical simulation and hardware prototyping. Proven experience in designing industrial machine vision systems and complex multiphoton setups. Specialized in wavefront shaping, **GPU-accelerated simulations**, and leveraging **LLMs** to optimize coding workflows and experimental control.

TECHNICAL SKILLS

Optical Design & Alignment

Multiphoton Microscopy (2P/3P), Adaptive Optics (SLM/Deformable Mirrors), Interferometry, Beam Shaping, Fiber Optics, Spectrometers.

High-Performance Computing

GPU Computing (CuPy, PyTorch for FFTs), **LLM Integration** (Copilot/GPT for code acceleration), Parallel Computing, Image Processing Pipelines.

Hardware & Automation

LabVIEW (FPGA Real-time), Industrial Cameras (Machine Vision), Motion Control, PMT/APD detection optimization.

Simulation & Analysis

Python (NumPy, SciPy), Optical Wave Propagation (Diffractive modeling), Phase Retrieval Algorithms, Fourier Optics.

Languages

English (Proficient), German (Speakable), Spanish (Native).

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Innsbruck, Austria

PhD Researcher - Adaptive Optics

Jan 2022 – Present

- Designed and built** a custom two-photon microscope capable of deep-tissue imaging via wavefront shaping.
- Accelerated simulation speed** by implementing **GPU-based algorithms** (CuPy) for complex wave propagation and phase retrieval.
- Developed novel algorithms** for correcting aberrations in scattering media, utilizing LLM tools to rapid-prototype Python control scripts.
- Automated the control of Spatial Light Modulators (SLM) and galvo-scanners using Python and FPGA.

École Polytechnique

Paris, France

R&D Optical Intern

Jul 2025 – Aug 2025

- Implemented variable Numerical Aperture (NA) control to optimize signal-to-noise ratio in nonlinear imaging systems.
- Engineered detection paths specifically for high-scattering environments.

Leibniz Institute of Photonic Technology (IPHT)

Assistant Researcher

Jena, Germany

Jun 2020 – Dec 2021

- Assembled setups for vibrational imaging (Raman/IR) applied to rapid cancer tissue diagnostics.
- Integrated photothermal microscopy hardware for environmental microplastic detection.

INTECOL S.A.S.

Machine Vision Engineer

Medellin, Colombia

Mar 2018 – Mar 2019

- ****Industrial Success:**** Designed and deployed quality control systems for production lines using high-speed cameras.
- Developed and deployed custom image processing algorithms for automated defect detection, reducing manual inspection error rates.
- Managed the full hardware lifecycle: from optical component selection to on-site installation and calibration.

EDUCATION

Medical University of Innsbruck

Ph.D. Advanced Microscopy

Innsbruck, Austria

Expected 2026

Friedrich Schiller University Jena

M.Sc. in Photonics

Jena, Germany

2019 – 2021

University of Cauca

B.Sc. in Physics Engineering

Colombia

2012 – 2018

SELECTED PUBLICATIONS & AWARDS

- **Awards:** ZEISS Summer School (2025), COLFUTURO Grant (2019).
- **J. D. Muñoz-Bolaños**, et al., "Scatter compensation for enabling deep two-photon microscopy using a MEMS phase light modulator," *Proc. SPIE* (2026).
- **J. D. Muñoz-Bolaños**, et al. "Wavefront corrections over large fields of view via beam cone tomography." *Proc. SPIE*, (2025).

REFERENCES

Prof. Monika Ritsch-Marte

Medical University of Innsbruck

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Prof. Jürgen Popp

Leibniz Institute of Photonic Technology

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